MODEL 3, MODEL Y, AND MODEL S / X FROM 2021 ON

Newer cars

Heat pump, control arms, phone keys, the small lithium battery, charging myths, and FSD.

October 2026 · The banner “Service is required” is useless without the alert name under the bell

No heat, no A/C, climate needs service

Model Y has had a heat pump from the start. Model 3 from about late 2020. A 2018 or 2019 Model 3 still has the older electric heater. Do not call that one a heat pump. On a heat-pump car the usual story is no heat in the cold, a loud compressor, a chemical smell, or the defrost giving up. Some 2021–2022 cars were in software recall 22V-063, a valve that stuck and killed the heat. Later failures are sensors, the valve block, or the compressor. They are not a dirty filter.

Ask whether they lost heat, A/C, or both, and whether it died while the cabin was preheating in the cold. A musty smell with the A/C still cold is usually the cabin filter and the evaporator, not a failed pump. The fan can run and the air can still be dead. Tell them that is expected for this fault.

Clunk or squeak in a parking lot

Worse when cold, especially over a bump with the wheel turned: front upper control arm or its mount. This is the most common Model 3 and Model Y complaint. It can be noisy before it is loose. Do not tell them it is “just the sound of an electric car.” Note left or right, cold only or always, and whether the car wanders. Inner-edge tire wear means an alignment at the same visit.

A 2025–2026 Model Y clunk over small bumps has its own service bulletins. Some 2026 Model Ys were recalled for front suspension fasteners, 26V558. Check the VIN before you call a noise ordinary wear.

Keys, pins, and stalkless cars

Newer cars use a phone and a card, not a fob. When the phone “stops working,” a phone update or a Bluetooth permission is more common than a broken car. Have them stand at the pillar, toggle Bluetooth, and try the card on the pillar. The card is the backup. It lives in the wallet, not in the car. PIN to Drive and a forgotten glovebox PIN are the same conversation as on the old cars: service can clear a forgotten PIN with proof of ownership. Do not guess.

A stalkless Highland or Juniper, and the refreshed S/X yoke, make people think the car will not start. The shifter is on the screen. The turn signals are on the wheel. Walk them through that before you book a dead car.

The small battery

Early Model 3s used a normal 12-volt lead-acid battery. From about 2022, and on refreshed S/X, it is a lithium low-voltage battery. Do not send them to a parts store. The car usually warns before it dies. Jump only from the posts in the manual, under the tow cover in the front bumper on a 3 or Y. The frunk will not open once that battery is flat.

Charging

Often not a failure

- Regen is weaker when the battery is full or cold.
- Supercharging slows as the battery fills. That is the curve, not a fault.
- A cold battery charges slowly until it warms.
- Scheduled charging sits there and looks broken until the time they set.

- A third-party charger or a loose adapter causes more “it will not charge” calls than a bad port.

Ask those before you write a charge-port ticket. A handle that is hot, a charge that fails on every stall, or an isolation warning during a charge is real. Note AC, DC, or both. Isolation codes and the open-versus-closed test are on the isolation desk card.

Cameras and FSD

SAY THIS FIRST

You are still the driver. “Take over immediately” means take over. Sunglasses, a hat, or a tag hanging over the cabin camera will nag them or shut assistance off. Sun in the front cameras does the same. Clean the glass before a service visit.

What they have	What to tell them
HW3, mostly cars built before early 2023	They get the lighter FSD, v14 Lite, not the full new stack. The computer can run warm. If Autopilot drops out until the car sits, note the temperature and that it returns after cooling. Unsupervised driving is not coming to this computer. Do not give a date for a replacement.
HW4 / AI4, newer 3, Y, S, and X	Full current FSD. Still supervised. A windshield replacement needs a camera calibration, or Autopilot will be wrong or unavailable. Put the glass date on the ticket.
No ultrasonic parking sensors, roughly late 2022 on	It will not beep the way an older car did. The picture on the screen is the parking aid. Do not book “backup sensors dead” on a car that never had them.

Water, smells, and the little stuff

- Model 3 trunk wet after rain. A known seal and lamp path. Note which side.
- Fog inside a headlamp that clears is condensation. A pool of water in the lamp is a leak.
- Sentry overnight, not plugged in, can use several percent. That is not a failing big battery.
- A phone that cooks on the wireless pad: take it off the pad. The pad is the problem.
- Seatbelt chime with an empty seat: the occupancy mat, not a haunted belt.
- Wipers on a dry windshield, or silent in the rain: show them the manual speed before you replace hardware.

A software update that will not finish

The small battery has to be healthy, the car has to be in Park, and the network has to be decent. Wi-Fi beats the cell connection. Tell them to leave it on the cord overnight and stop tapping retry. If it has been stuck for days, or the screen boot-loops, then it is a ticket. Write how long it has been stuck and whether the car still drives.

Refreshed S and X air suspension

Same owner words as the old cars, different parts. One corner down after sitting is still a leak. A replacement control module still has to be programmed, and the height still has to be calibrated. A yoke and a tilting screen are not suspension problems. Do not let those ride along on the ticket.

They scraped the bottom

If the car drives normally, with no smell and no new warning, it can be inspected. If they smell coolant, see fluid, hear a change, or get an isolation or battery warning, they stop. Do not rinse the underside and hope. Write the speed, and whether it was a speed bump, a curb, or road debris.

STOP AND TOW

Smoke or a chemical smell from the heater. An isolation or battery warning. A wheel that looks out of place. Steering that suddenly goes heavy. A charge handle that is melting, or too hot to hold.

THE TICKET LINE

Model and year. Heat pump or not. HW3 or HW4. The exact alert under the bell. Left or right. Cold or always. AC or Supercharger. Whether a windshield was replaced recently.